📘 E-Commerce Data Warehouse (ECOM_DEV_DB)

Complete Technical Documentation

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1. Introduction

The **E-Commerce Data Warehouse (ECOM_DEV_DB)** is a production-style SQL data warehouse designed to transform raw operational data into trusted analytical datasets.

The project follows a **layered ETL architecture** with integrated **Data Quality (DQ)** validation, **Audit Logging**, **Reference Data Management**, **Dimensional Modeling**, and **Business Analytics**.

The objective is to demonstrate a complete end-to-end data engineering workflow using SQL while following industry-standard data warehousing practices.

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2. Project Objectives

The project is designed to:

- Build a scalable layered warehouse
- Separate ingestion, cleansing and reporting
- Validate data quality before each transformation
- Maintain audit history of every pipeline execution
- Centralize business logic in reusable SQL views

• Provide trusted analytical datasets for reporting
• Demonstrate production-style ETL architecture

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3. Architecture Overview

```
                 Source Files
                      │
                      ▼
               ┌─────────────┐
               │     RAW     │
               └─────────────┘
                      │
                      ▼
          RAW VALIDATION (Views)
                      │
                      ▼
               ┌─────────────┐
               │    STAGE    │
               └─────────────┘
                      │
                      ▼
        STAGE VALIDATION (Views)
                      │
                      ▼
               ┌─────────────┐
               │    CLEAN    │
               └─────────────┘
                      │
                      ▼
       CLEAN VALIDATION (Views)
                      │
                      ▼
               ┌─────────────┐
               │    GOLD     │
               └─────────────┘
                      │
                      ▼
             Analytics SQL Views

         Supporting Layers

         • Data Quality
         • Audit
         • Reference
```

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4. End-to-End Data Flow

The complete pipeline follows the sequence:

```
RAW
  ↓
RAW TEST
  ↓
STAGE
  ↓
STAGE TEST
  ↓
CLEAN
  ↓
CLEAN TEST
  ↓
GOLD
  ↓
ANALYTICS
```

Supporting layers operate throughout the pipeline:

- Data Quality (DQ)
- Audit
- Reference

This design ensures only validated data progresses through each stage.

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5. Layer Architecture

RAW Layer

Purpose

Stores source data exactly as received.

Tables

- CUSTOMERS_RAW
- ORDERS_RAW
- PRODUCTS_RAW
- EMPLOYEES_RAW
- PAYMENTS_RAW
- SHIPMENTS_RAW

Responsibilities

- Data ingestion
- Preserve source records
- Minimal transformation

Validation

RAW validation views perform:

- NULL checks
- Blank value checks
- Invalid numeric values
- Basic format validation
- Mandatory field validation

Only validated records move to STAGE.

STAGE Layer

Purpose

Standardizes raw data into a consistent structure.

Tables

- CUSTOMERS_STAGE
- ORDERS_STAGE
- PRODUCTS_STAGE
- EMPLOYEES_STAGE
- PAYMENTS_STAGE
- SHIPMENTS_STAGE

Responsibilities

- Standardization
- Data normalization
- Business rule enforcement

Validation

Stage validation includes:

- Duplicate detection
- Business rule validation
- Invalid value detection
- Consistency checks

Validated data proceeds to CLEAN.

CLEAN Layer

Purpose

Trusted business-ready datasets.

Tables

- CUSTOMERS_CLEAN
- ORDERS_CLEAN
- PRODUCTS_CLEAN
- EMPLOYEES_CLEAN
- PAYMENTS_CLEAN
- SHIPMENTS_CLEAN

Responsibilities

- Final cleansing
- Reliable reporting data
- Standardized business entities

Validation

- Data completeness
- Business rule compliance
- Consistency validation
- Final quality verification

Validated data is loaded into GOLD.

GOLD Layer

Purpose

Dimensional model optimized for analytics.

Dimension Tables

- DIM_CUSTOMERS
- DIM_PRODUCTS
- DIM_EMPLOYEES
- DIM_DATE

Fact Tables

- FACT_ORDERS
- FACT_PAYMENTS
- FACT_SHIPMENTS

The GOLD schema serves as the analytical foundation for reporting.

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6. Data Quality Framework

The project includes dedicated SQL validation views between every processing layer.

Current validation categories include:

- NULL value checks
- Blank string checks
- Negative numeric value checks
- Basic format validation
- Duplicate detection
- Business rule validation
- Data consistency checks

Validation logic is separated from transformation logic, making the ETL process easier to maintain and troubleshoot.

Rejected records are redirected to the **DQ Layer** for investigation.

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7. Analytics Layer

Overview

The Analytics schema contains **57 business-oriented SQL views** built on the GOLD star schema. These views centralize business logic and provide reusable datasets for dashboards, reporting, and business analysis.

Sales & Revenue Analytics (10 Views)

Views

- VW_TOP_PRODUCTS_ANALYTICS
- VW_MONTHLY_REVENUE_ANALYTICS
- VW_REVENUE_BY_CATEGORY_ANALYTICS
- VW_REVENUE_BY_CURRENCY_ANALYTICS
- VW_REVENUE_BY_COUNTRY_ANALYTICS
- VW_AVG_ORDER_VALUE_ANALYTICS
- VW_SALES_BY_EMPLOYEE_ANALYTICS
- VW_FASTEST_GROWING_COUNTRIES_ANALYTICS
- VW_SEASONAL_SALES_ANALYTICS
- VW_PROMOTION_IMPACT_ANALYTICS

Business Insights

- Revenue trends
- Top-performing products

- Sales by geography
- Employee performance
- Seasonal sales behavior

Customer Analytics (12 Views)

Views

- VW_TOP_CUSTOMERS_ANALYTICS
- VW_NEW_CUSTOMERS_MONTHLY_ANALYTICS
- VW_REPEAT_PURCHASE_RATE_ANALYTICS
- VW_CUSTOMER_CHURN_ANALYTICS
- VW_CUSTOMER_SEGMENTATION_ANALYTICS
- VW_AVG_CUSTOMER_SPEND_ANALYTICS
- VW_CUSTOMER_ACQ_COST_ANALYTICS
- VW_CUSTOMER_RETENTION_ANALYTICS
- VW_CUSTOMER_AGE_GROUP_ANALYTICS
- VW_VIP_CUSTOMERS_ANALYTICS
- VW_CUSTOMER_LIFETIME_VALUE_ANALYTICS
- VW_CUSTOMER_COHORT_ANALYTICS

Business Insights

- Customer lifetime value
- Churn analysis
- Customer retention
- Customer segmentation
- VIP customer identification

Orders & Fulfillment Analytics (12 Views)

Views

- VW_ORDER_CANCELLATION_ANALYTICS
- VW_LATE_DELIVERIES_ANALYTICS
- VW_DELIVERY_TIME_BY_CARRIER_ANALYTICS
- VW_ORDERS_BY_PAYMENT_METHOD_ANALYTICS
- VW_ORDERS_BY_CATEGORY_ANALYTICS
- VW_ORDERS_BY_COUNTRY_MONTHLY_ANALYTICS
- VW_BASKET_ANALYSIS_ANALYTICS
- VW_RETURN_RATE_BY_CATEGORY_ANALYTICS
- VW_PAYMENT_MISMATCH_ANALYTICS
- VW_ORDER_OUTLIERS_ANALYTICS
- VW_DELIVERY_SLA_COMPLIANCE_ANALYTICS
- VW_RETURN_REASON_ANALYTICS

Business Insights

- Delivery performance
- SLA compliance
- Basket analysis
- Return analysis
- Payment reconciliation

Inventory & Supplier Analytics (10 Views)

Views

- VW_OUT_OF_STOCK_PRODUCTS_ANALYTICS
- VW_HIGH_MARGIN_PRODUCTS_ANALYTICS
- VW_STOCK_TURNOVER_ANALYTICS
- VW_SUPPLIER_PERFORMANCE_ANALYTICS
- VW_PRODUCT_POPULARITY_ANALYTICS
- VW_DECLINING_PRODUCTS_ANALYTICS
- VW_HIGH_RETURN_PRODUCTS_ANALYTICS
- VW_INVENTORY_AGING_ANALYTICS
- VW_TOP_SUPPLIERS_ANALYTICS
- VW_PRODUCT_LIFECYCLE_ANALYTICS

Business Insights

- Inventory health
- Supplier performance
- Product popularity
- Margin analysis
- Product lifecycle

Finance & Operational Analytics (13 Views)

Views

- VW_PAYMENT_SUCCESS_RATE_ANALYTICS
- VW_CURRENCY_EXCHANGE_IMPACT_ANALYTICS
- VW_FRAUD_DETECTION_ANALYTICS
- VW_LOGISTICS_COST_ANALYTICS
- VW_EMPLOYEE_PRODUCTIVITY_ANALYTICS
- VW_COUNTRY_PROFITABILITY_ANALYTICS
- VW_CAMPAIGN_ROI_ANALYTICS
- VW_OPERATIONAL_EFFICIENCY_ANALYTICS
- VW_REFUND_PROCESSING_TIME_ANALYTICS
- VW_COMPLIANCE_CHECKS_ANALYTICS
- VW_CHANNEL_PERFORMANCE_ANALYTICS
- VW_CAMPAIGN_CONVERSION_ANALYTICS
- VW_SUSTAINABILITY_IMPACT_ANALYTICS

Business Insights

- Operational efficiency
- Logistics cost
- Fraud detection
- Profitability
- Marketing performance

Analytics Summary

Domain	Views
Sales & Revenue	10
Customer Analytics	12
Orders & Fulfillment	12
Inventory & Supplier	10
Finance & Operations	13
Total Analytics Views	**57**

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8. Audit Framework

The Audit layer records operational information for every pipeline execution.

Tables include:

- AUDIT_PIPELINE_RUNS
- AUDIT_TABLE_LOADS
- AUDIT_SQL_QUERY_LOG
- AUDIT_FILE_INGESTION

Captured information includes:

- Pipeline status
- Execution time
- Loaded records
- Rejected records
- SQL execution history
- File ingestion details

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9. Reference Data Layer

The Reference schema contains master data used across the warehouse.

Examples include:

- Countries
- Currency
- Product Categories
- Suppliers
- Order Status
- Shipment Status
- Payment Methods
- Departments
- Employee Roles

Reference data ensures consistency throughout the ETL process.

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10. Pipeline Execution

Execution order:

1. Load source data into RAW
2. Execute RAW validation views
3. Load validated data into STAGE
4. Execute STAGE validation views
5. Load validated data into CLEAN
6. Execute CLEAN validation views
7. Load dimensional model into GOLD
8. Execute Analytics views
9. Review DQ records
10. Review Audit logs

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11. Project Deliverables

The project includes:

- Layered SQL Data Warehouse
- ETL SQL Scripts
- Data Validation Views
- Star Schema
- Analytics Views (57)
- Data Quality Framework
- Audit Framework
- Reference Data
- Architecture Diagrams
- Technical Documentation

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12. Business Value

The project demonstrates how structured data engineering practices improve analytical reliability.

Business benefits include:

- Trusted analytical datasets
- High data quality through staged validation
- Complete auditability
- Centralized business logic
- Reusable reporting views
- Modular architecture
- Scalable warehouse design

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13. Technology Stack

Component	Technology
Cloud Data Warehouse	Snowflake
Language	SQL
Data Modeling	Star Schema
ETL	SQL-based ETL Pipeline
Data Quality	SQL Validation Views
Documentation	Technical Documentation & Architecture Diagrams

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14. Future Enhancements

Potential future improvements include:

- Workflow orchestration using Apache Airflow
- Incremental loading with Streams & Tasks
- Slowly Changing Dimensions (SCD Type 2)
- CI/CD integration
- Automated data quality alerts
- Dashboard integration (Power BI / Tableau)

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15. Conclusion

The **E-Commerce Data Warehouse (ECOM_DEV_DB)** demonstrates a complete production-style SQL data warehouse implementation. The project integrates layered ETL processing, data quality validation, audit logging, reference data management, dimensional modeling, and a comprehensive analytics layer consisting of **57 business-focused SQL views**.

The solution follows industry-standard data warehousing principles and provides a maintainable, scalable, and business-oriented architecture suitable for analytical reporting and decision support. It showcases practical SQL development, data engineering concepts, and business intelligence fundamentals in a single end-to-end portfolio project.